

Day 14 – AWS Cloud Security & Encryption

Introduction

Cloud security protects infrastructure, applications, and data.

Security Principles

Confidentiality, Integrity, Availability.

AWS Security Services

IAM, KMS, CloudTrail, GuardDuty, Shield.

Encryption

Encryption converts readable data into ciphertext.

Encryption Types

At Rest and In Transit encryption.

AWS KMS

Manages encryption keys securely.

Project

Secure File Storage System using S3 + KMS + CloudTrail.

Monitoring

CloudTrail logs activity and GuardDuty detects threats.

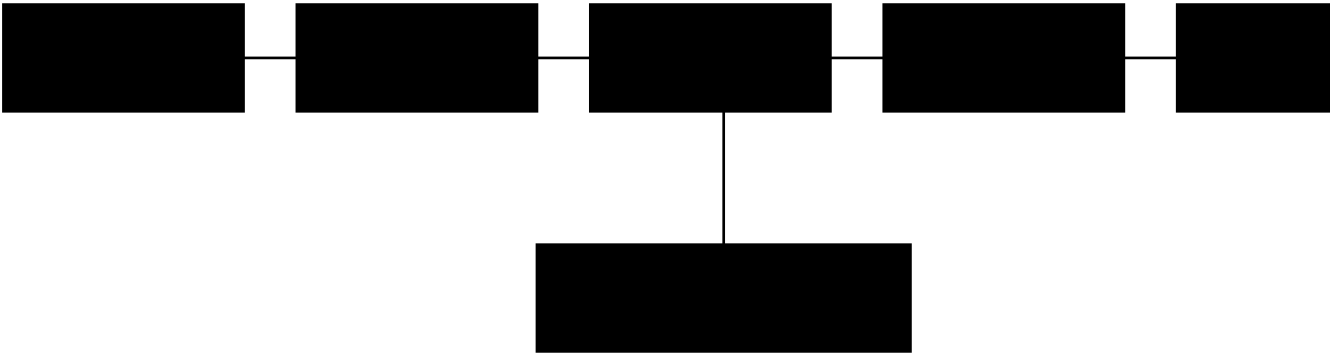
Use Cases

Banking, Healthcare, AI security, Enterprise cloud security.

Hands-On

Enable encryption, configure IAM, enable CloudTrail.

Security Workflow Diagram



AWS Security Services

Service	Purpose
IAM	Access Control
KMS	Encryption Management
CloudTrail	Logging & Auditing
GuardDuty	Threat Detection

Encryption Workflow

User Data ↓ AWS KMS Encrypts Data ↓ Encrypted Data Stored in S3 ↓ Authorized User Access

Author: Sankar S – Cloud & AI Learning Journey